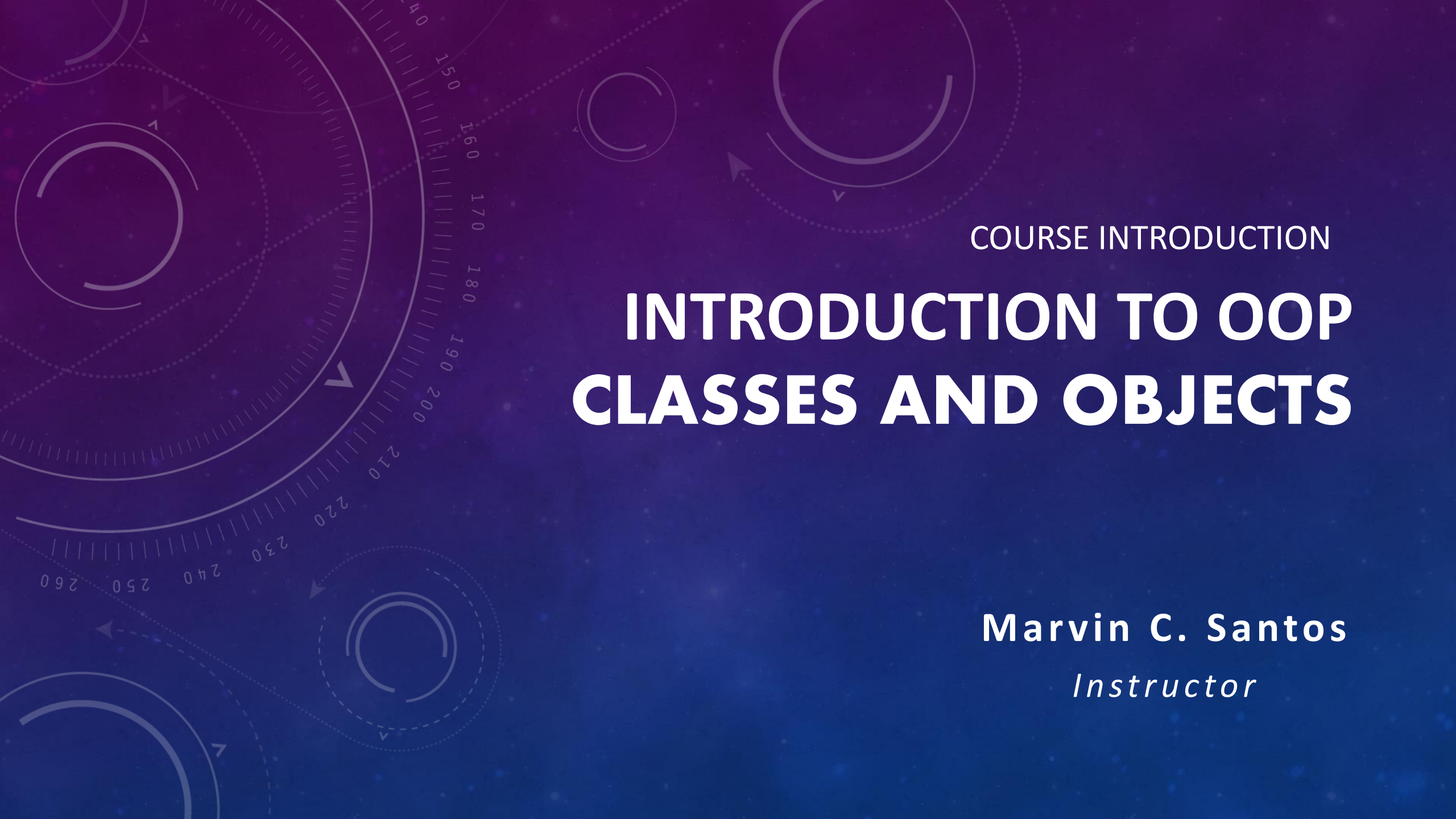


The background is a dark blue gradient with a subtle pattern of white dots. Overlaid on this are several faint, light blue circular elements. On the left side, there are large concentric circles with degree markings ranging from 40 to 260. On the right side, there are smaller concentric circles, some with arrows indicating a clockwise direction. The overall aesthetic is technical and modern.

COURSE INTRODUCTION

# INTRODUCTION TO OOP CLASSES AND OBJECTS

**Marvin C. Santos**

*Instructor*

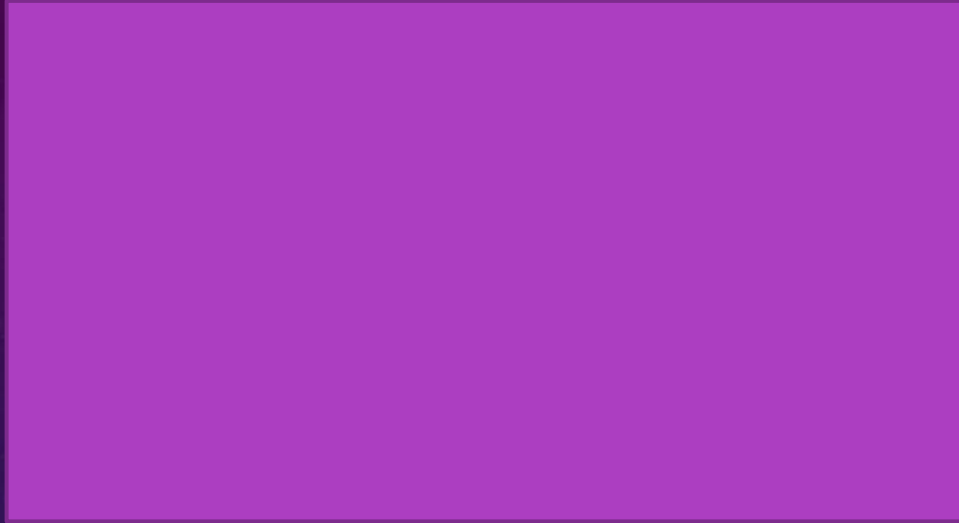
# ACTIVITY 1

Assume you undertake the following activities on a daily basis.

Arrange this list into a sensible order then split this into three blocks of related activities and give each block a heading to summaries the activities carried out in that block.

- Park the Car
- Get Dressed
- Find out what your boss wants you to do today
- Get the Car out of the garage
- Eat breakfast
- Feedback to the boss on today's result
- Get out of bed
- Do what the boss wants you to do
- Drive to work

# FEEDBACK 1



# FEEDBACK 1

**Get Up**

**Go to Work**

**Do your Job**

## FEEDBACK 1

# Programming Paradigm

### Get Up

Get out of bed  
Get Dressed  
Eat breakfast

### Go to Work

Get the Car out of the garage  
Drive to work  
Park the Car

### Do your Job

Find out what your boss wants you to do today  
Do what the boss wants you to do  
Feedback to the boss on today's result

# OOP Paradigm

- Suggest we should model instructions in a computer program with the data they manipulate and store these as components together.
- One advantage of doing this is we get reusable software components.

BEFORE OOP:

## PROCEDURAL PROGRAMMING

- Sequences of instructions to be executed
  - Splitting of programs into sets of instructions
- Divide a program into a SET of FUNCTIONS
  - Data Stored (Variables)
  - Functions that operates the data

# AS PROGRAMS GROWS: SPAGHETTI CODES:

- PROGRAM JUMPS ALL OVER THE PLACE  
AND IS DIFFICULT TO FOLLOW
- LOTS OF GO TO STATEMENTS



```
1 C A weird program for calculating Pi written in Fortran.
2 C From: Fink, D.G., Computers and the Human Mind, Anchor Books, 1966.
3
4 PROGRAM PI
5 DIMENSION TERM(100)
6 N=1
7 3 TERM(N)=((-1)**(N+1))*(4./(2.*N-1.))
8 N=N+1
9 IF (N-101) 3,6,6
10 6 N=1
11 7 SUM98 = SUM98+TERM(N)
12 WRITE(*,28) N, TERM(N)
13 N=N+1
14 IF (N-99) 7, 11, 11
15 11 SUM99=SUM98+TERM(N)
16 SUM100=SUM99+TERM(N+1)
17 IF (SUM98-3.141592) 14,23,23
18 14 IF (SUM99-3.141592) 23,23,15
19 15 IF (SUM100-3.141592) 16,23,23
20 16 AV89=(SUM98+SUM99)/2.
21 AV90=(SUM99+SUM100)/2.
22 COMANS=(AV89+AV90)/2.
23 IF (COMANS-3.1415920) 21,19,19
24 19 IF (COMANS-3.1415930) 20,21,21
25 20 WRITE(*,26)
26 GO TO 22
27 21 WRITE(*,27) COMANS
28 STOP
29 23 WRITE(*,25)
30 GO TO 22
31 25 FORMAT('ERROR IN MAGNITUDE OF SUM')
32 26 FORMAT('PROBLEM SOLVED')
33 27 FORMAT('PROBLEM UNSOLVED', F14.6)
34 28 FORMAT(I3, F14.6)
35 END
36
```

# OOP: TO THE RESCUE

COMBINE RELATED VARIABLES AND FUNCTIONS INTO A UNIT

F() X

## CAMERA

Model Name  
Code  
Year Manufactured  
Maker  
OS  
Memory

3D Enabled Camera( )  
LTE ( )  
GPS( )  
....  
....

# High Level Programming Language

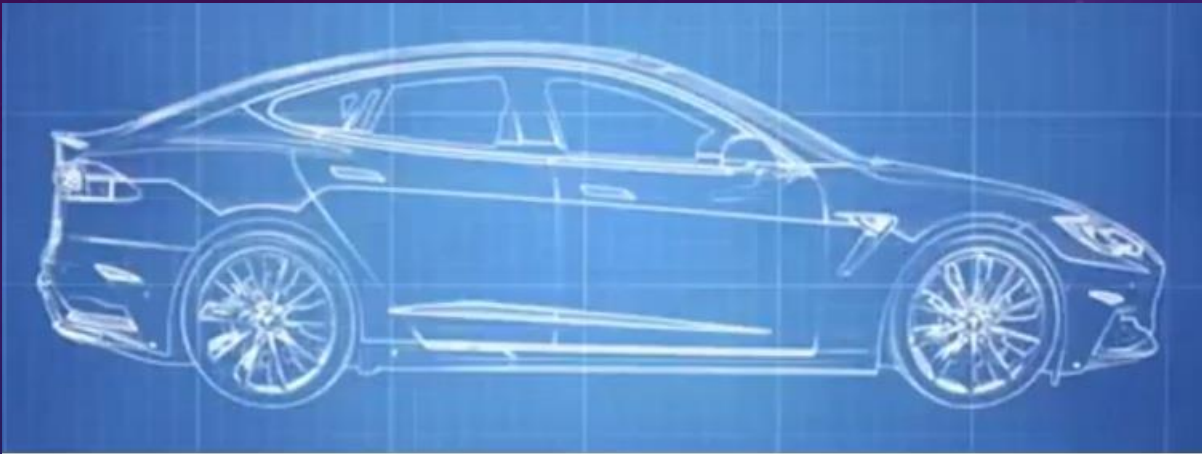
- **Java** - "write once, run anywhere"
- **C++** - extension of the C with OOP
- **JavaScript** - allows interactive and dynamic features added to a websites
- **Swift** - by Apple, used for building applications on Apple's platforms
- **PHP** - used for web development
- **Python** - Known for its simplicity and readability, Python is widely used in various domains, including web development, data analysis, scientific computing, and artificial intelligence.
- **C#** - by Microsoft, C# is used for building Windows applications

*"These languages vary in terms of their strengths, weaknesses, and areas of application."*

*"The choice of language depends on the specific requirements of a project and the preferences of the programmer or development team"*

# CLASSES AND OBJECTS

## CLASS



## OBJECTS



# CLASSES AND OBJECTS

## CLASS

- Is a *container* that has data that called attributes and functions that called methods

CLASS = CAR

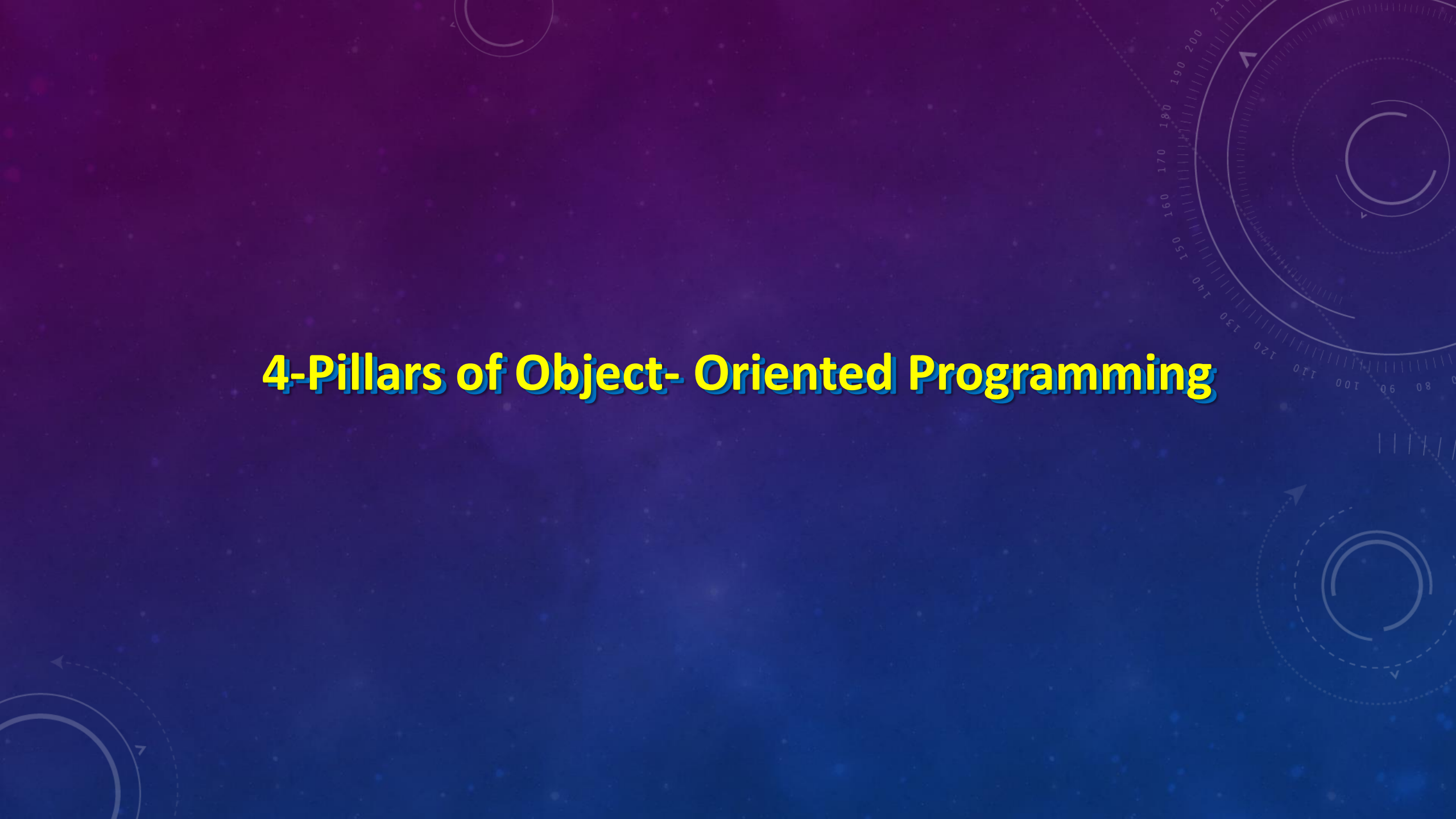
Attributes (Data)

Make, Color, Model

Functions (Methods)

start( ), move ( )...

# 4-Pillars of Object- Oriented Programming

The background is a dark blue gradient with a subtle pattern of white dots. Faint, light blue circular patterns are visible, including a large one in the top right corner with concentric circles and radial lines, and another in the bottom right corner with dashed lines and arrows. A small circular pattern is also visible in the top left corner.

# Encapsulation

- *bundling of data and methods into a single unit*

***Property + Method***



# HTML – WEB DESIGNS

- Eliminate redundant codes
- Defined Properties and Methods as:  
OBJECTS

## Inheritance

**{CSS}**

**strong** { color: red; }

**.chaptertext** { font-family: georgia; }

**<HTML>**

Georgia }

**<div class="chaptertext">**



inherit

red (bold) }

**<strong>**



inherit

(italics) }

**<em>**

Curiouser and curiouser!

**</em>**

**</strong>**

**</div>**

**Results**

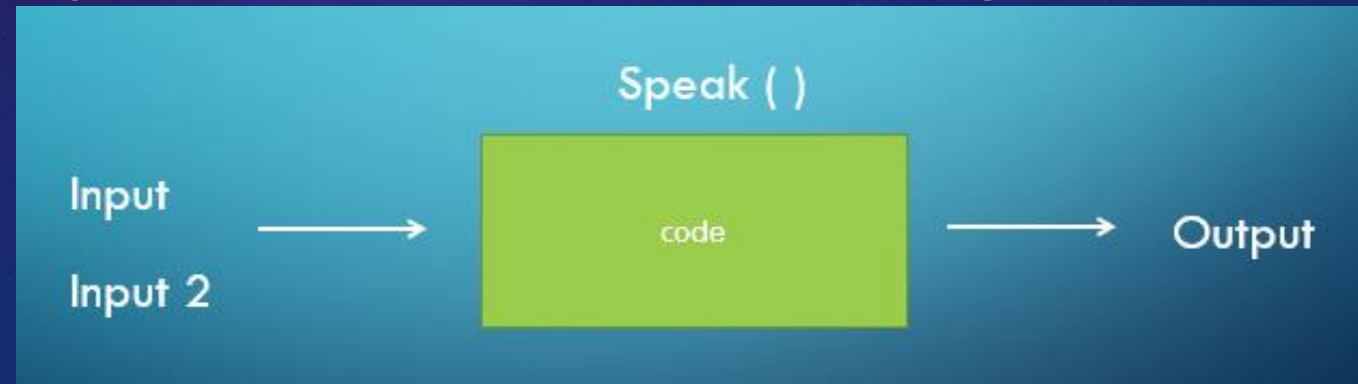
***Curiouser and curiouser!***

# Polymorphism

- *Many forms*
- *Achieved through the use of inheritance and method overriding*



***Poly + Morph***



# GAMING CONSOLE

- Complexity is Hidden
- Hides Property and Methods
  - Simpler Interface
  - Reduce the Impact of Change

## Abstraction



## ADVANTAGES OF OOP:

- Modularity
- Reusability
- Flexibility
- Encapsulation of Data
- Maintenance



THANK YOU!